

AUTH-CAT Analysis Report

Project: Tyrnavos
Generated: 2026-09-24 19:57

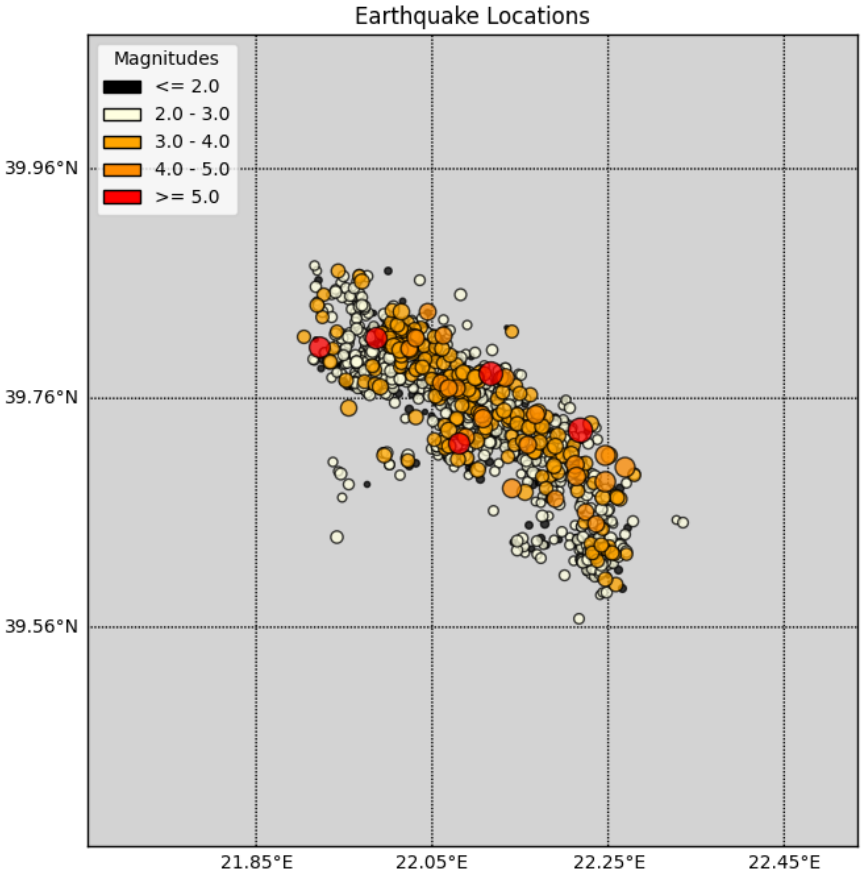
Dataset overview

This section summarizes the catalogue information stored in the AUTH-CAT analysis summary. Missing values are shown as '-'.

Project	Tyrnavos
Catalogue	Results/Tyrnavos_cat.txt
Download period	2021-03-03 - 2021-04-03
Download area	Lat: 39.5 - 39.9 deg; Lon: 21.85 - 22.35 deg
Number of events	1753
Time period	2021-03-03 08:00:37.674 - 2021-04-03 23:52:26.598
Study area	Lat: 39.5682 - 39.87683 deg; Lon: 21.85791 - 22.33374 deg
Depth range	0.049 - 26.455 km
Magnitude range	0.3-6.0

Spatial distribution

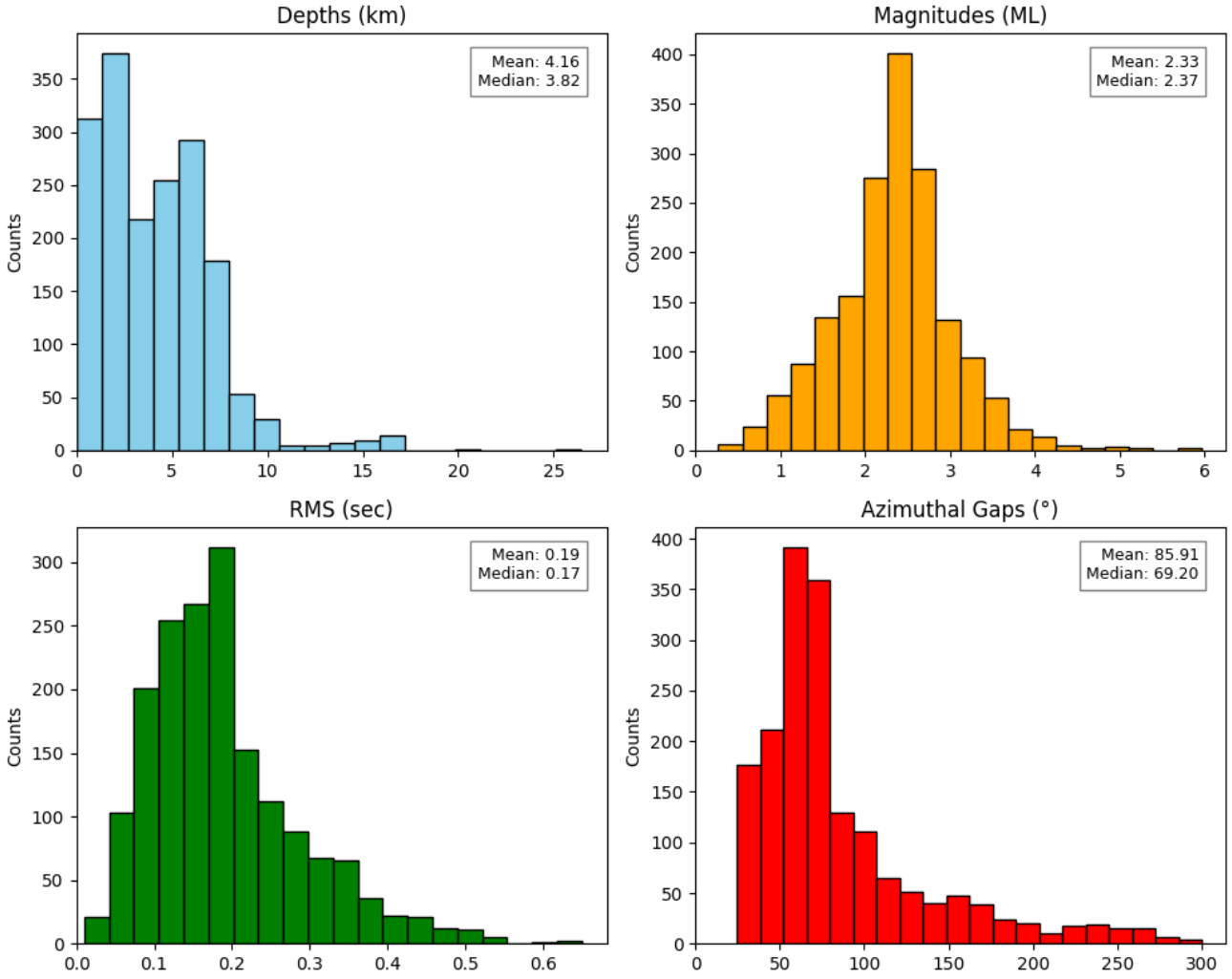
The epicentral map shows the spatial distribution of the events included in the catalogue.



Epicentral distribution of earthquakes.

Catalogue characteristics

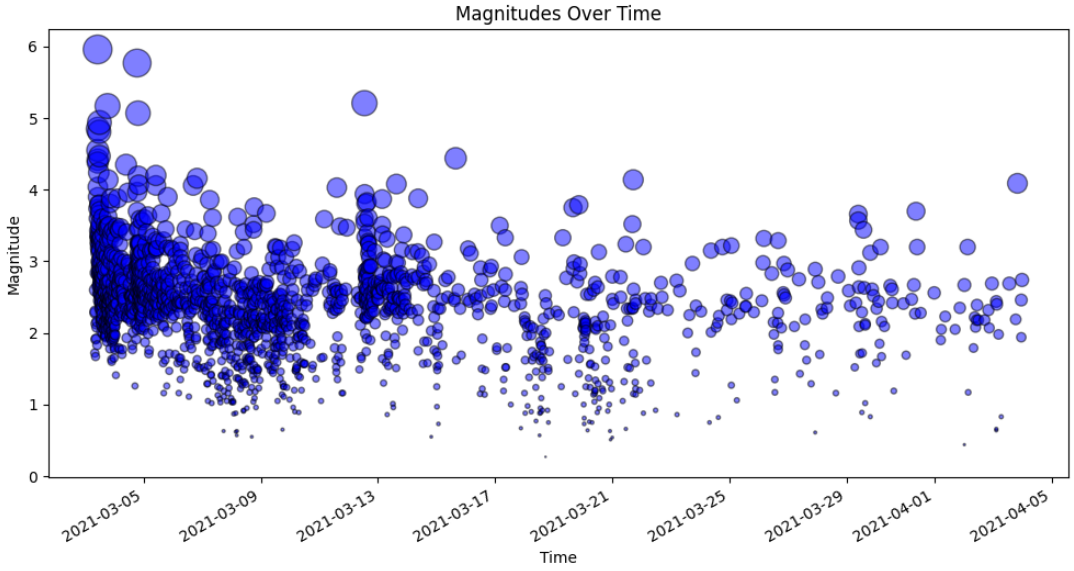
The catalogue histograms summarize parameters such as magnitude, focal depth, RMS and azimuthal gap, depending on the generated figure.



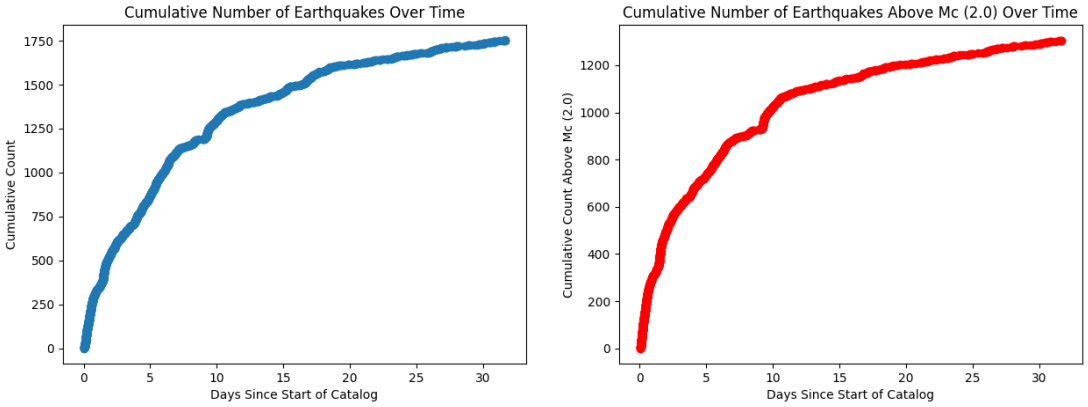
Catalogue parameter histograms.

Temporal evolution

These figures summarize the temporal development of the earthquake sequence using the outputs generated by AUTH-CAT.



Magnitude versus time.



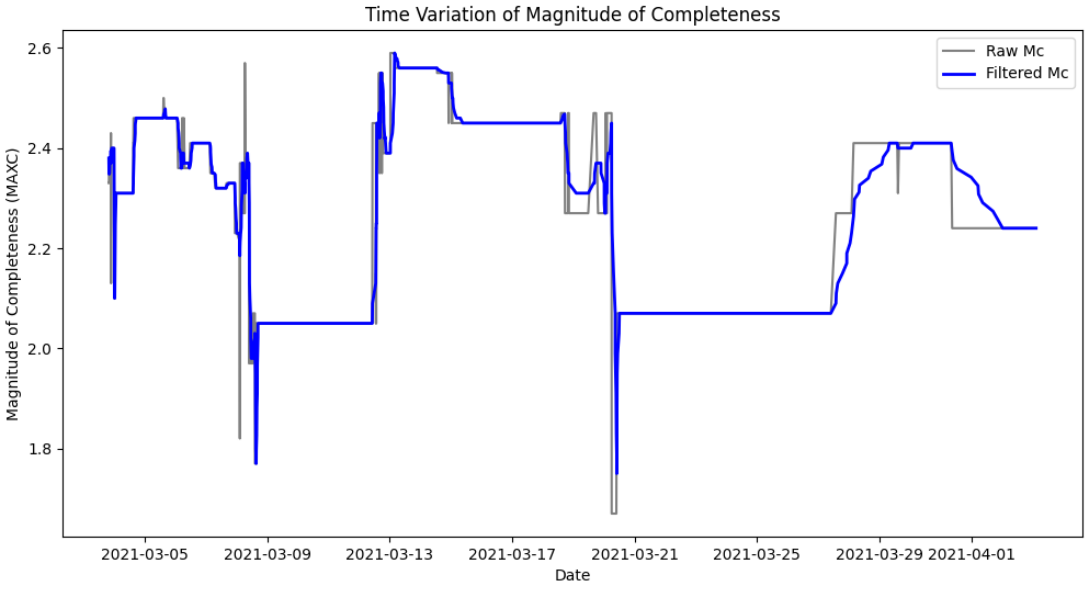
Cumulative number of earthquakes over time.

Magnitude of completeness

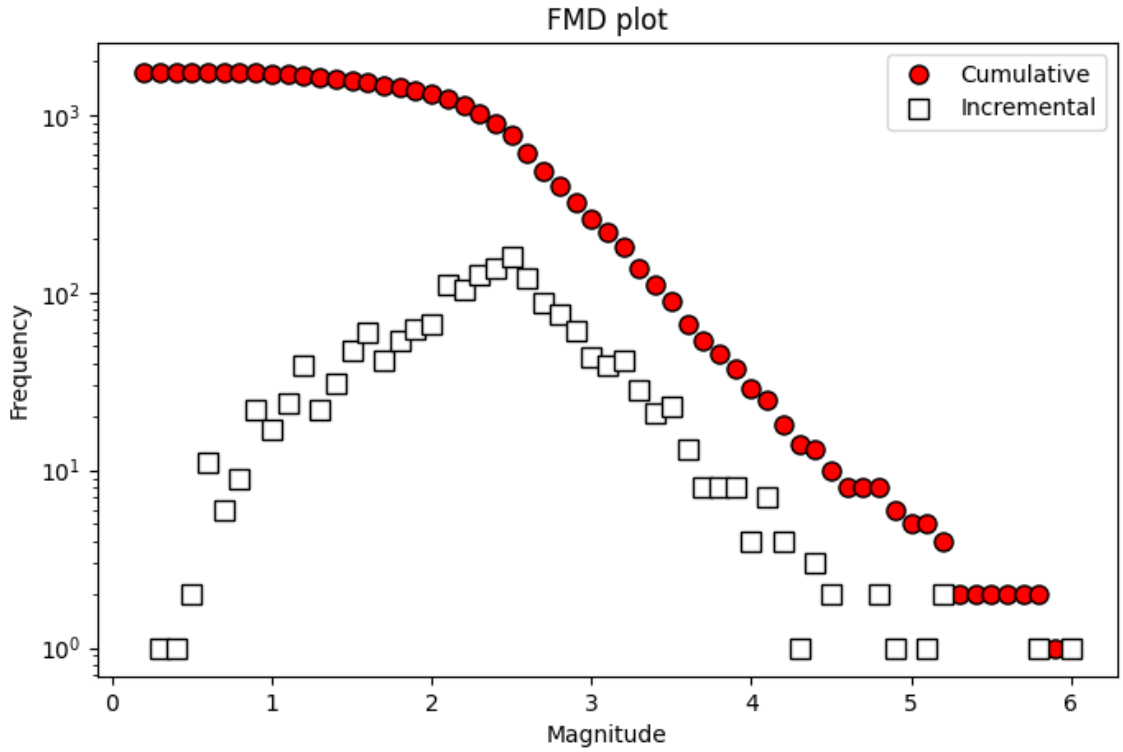
Magnitude of completeness (Mc) and Gutenberg-Richter b-values are reported for the methods stored in the analysis summary.

Method	Mc	b-value	b-value uncertainty
MAXC	2.5	0.93	0.03
GFT	2.4	0.89	0.03
MBS	2.4	0.89	0.03

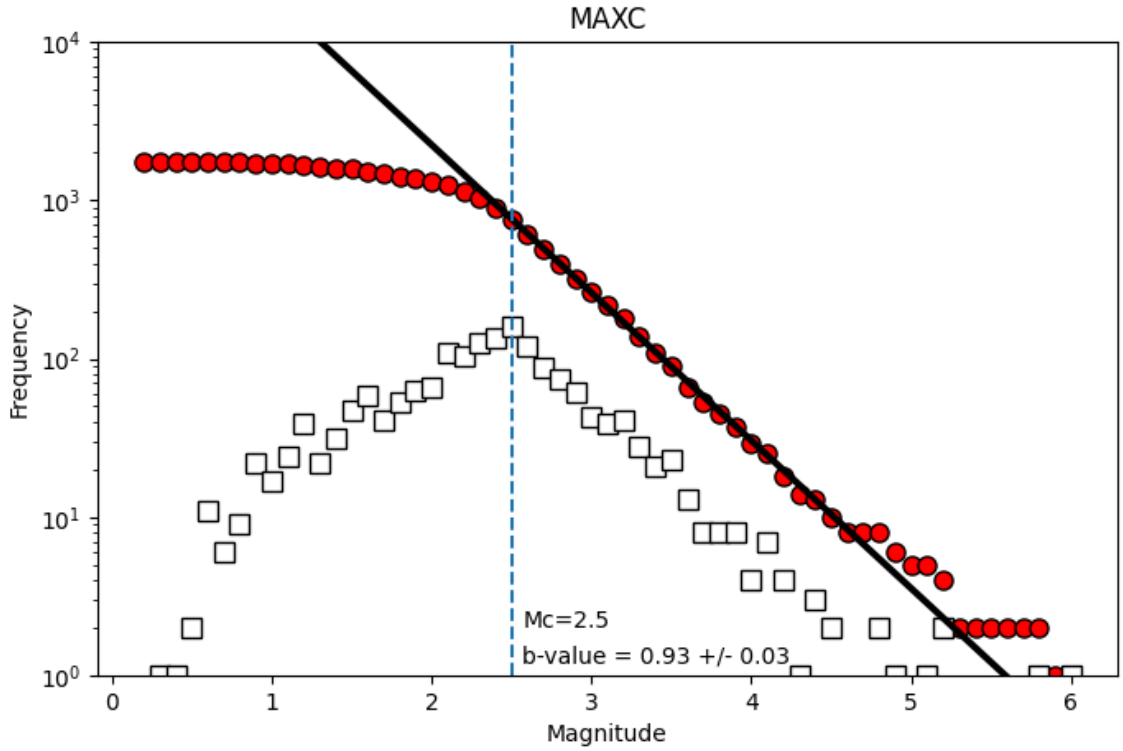
GFT goodness of fit: 95%



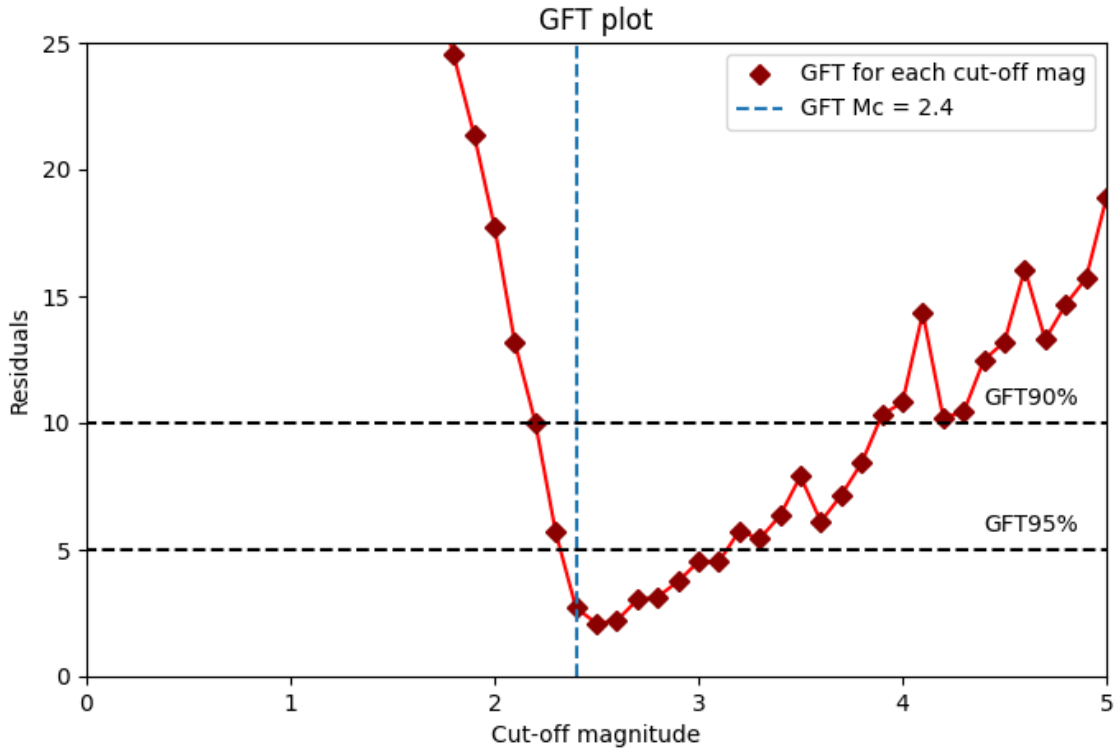
Magnitude of completeness through time.



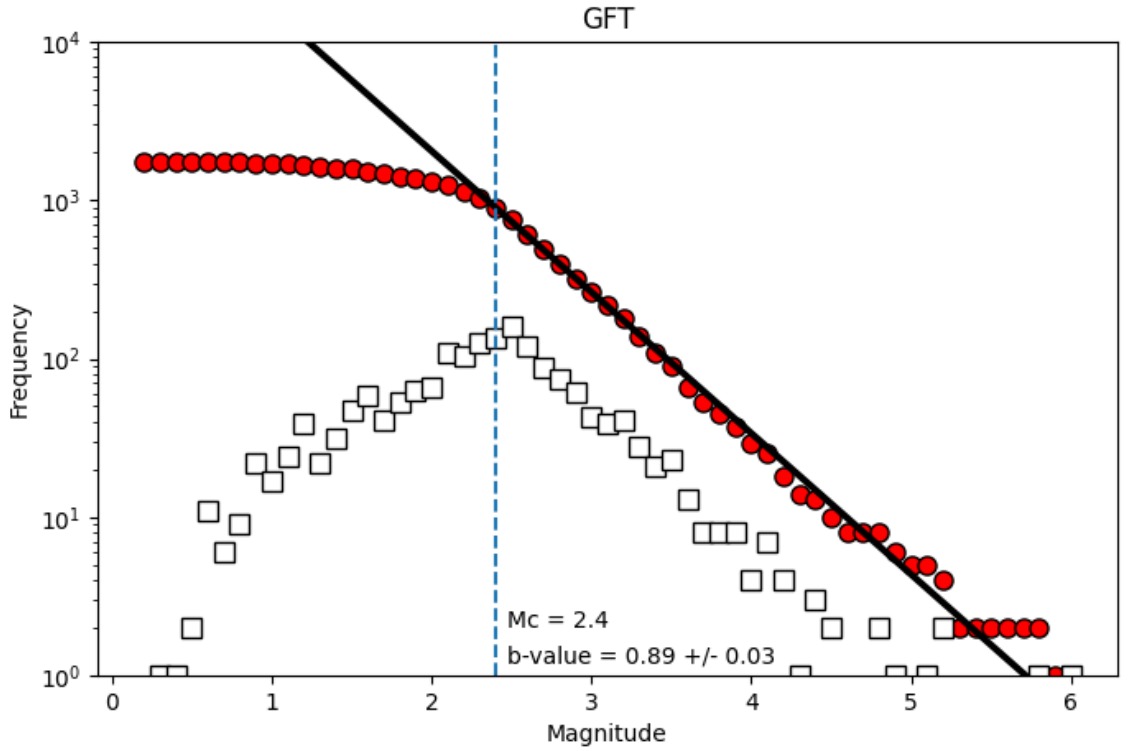
Frequency-magnitude distribution.



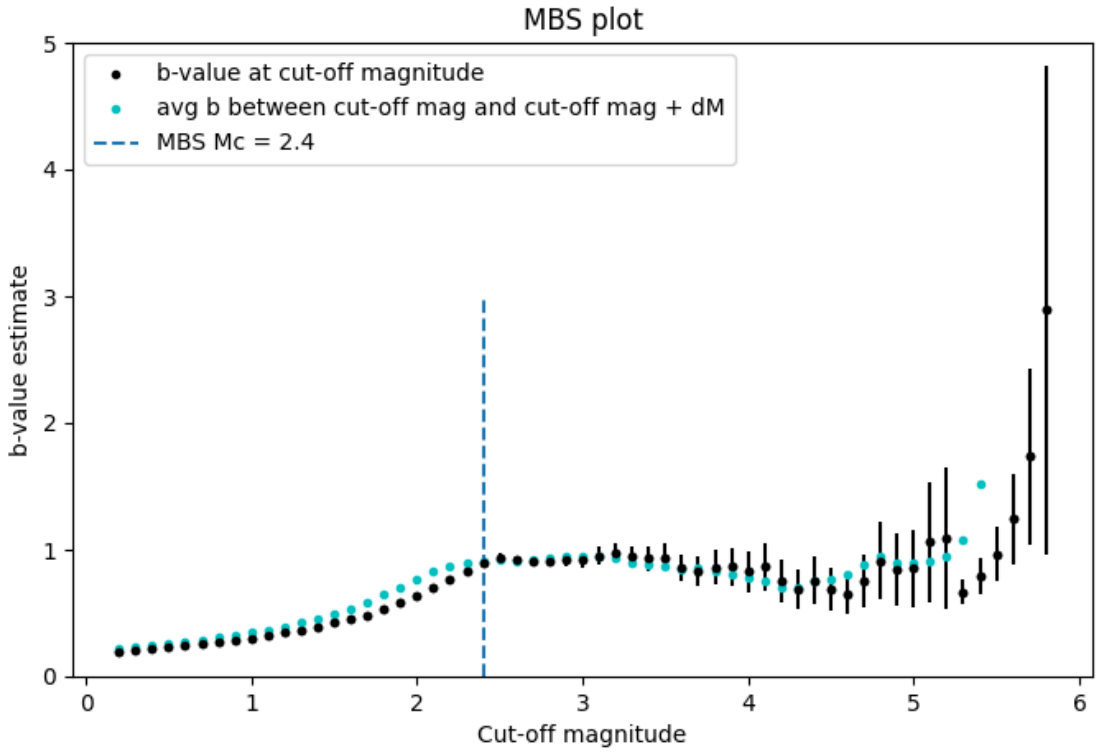
MAXC magnitude of completeness analysis.



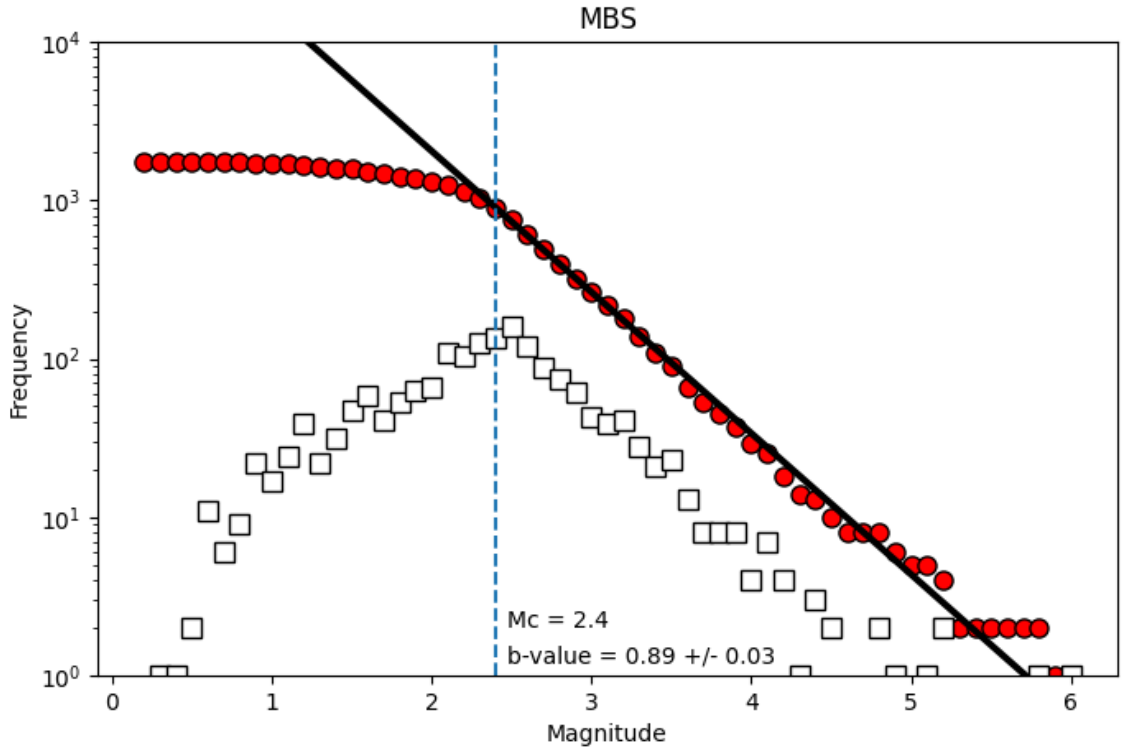
Goodness-of-Fit residuals.



Goodness-of-Fit magnitude of completeness analysis.



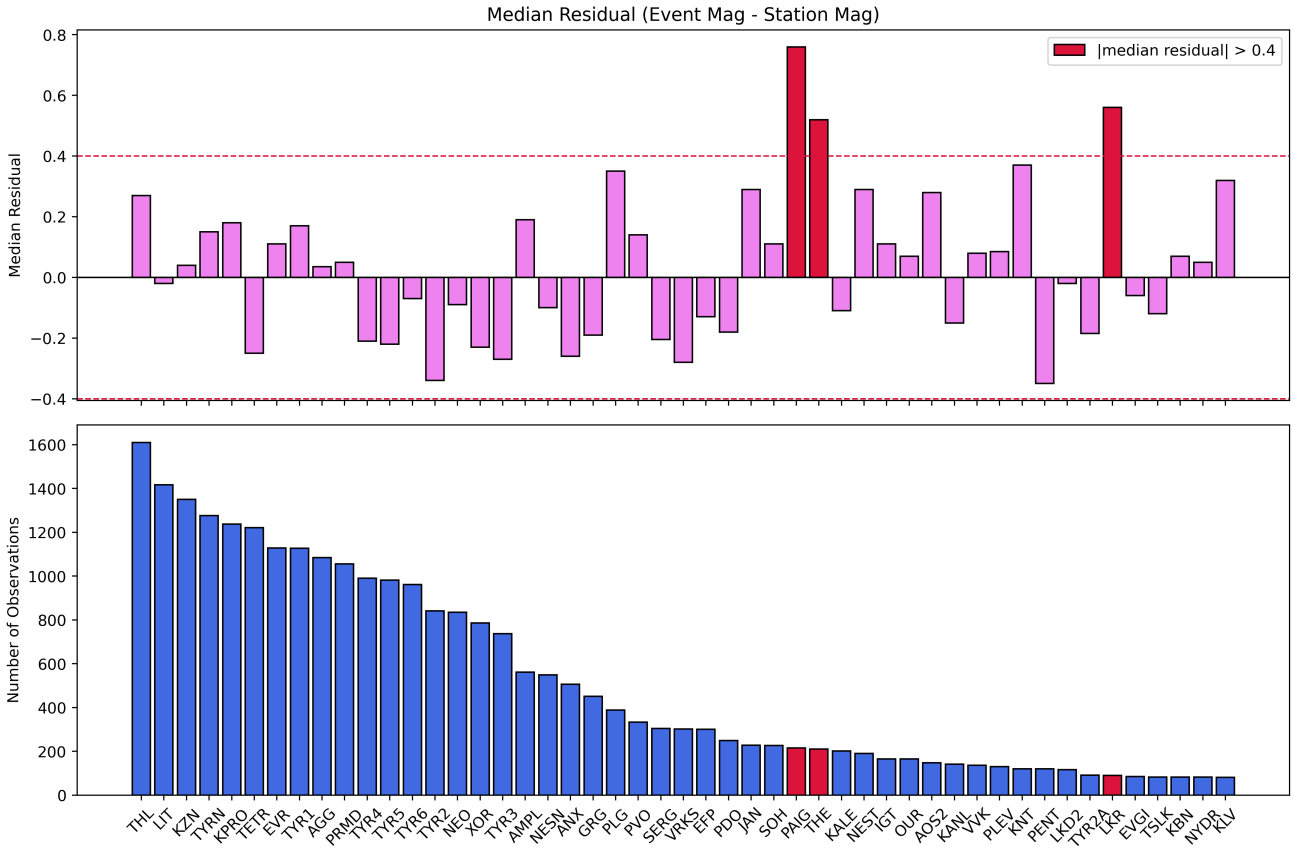
b-value stability analysis.



MBS magnitude of completeness analysis.

Magnitude analysis

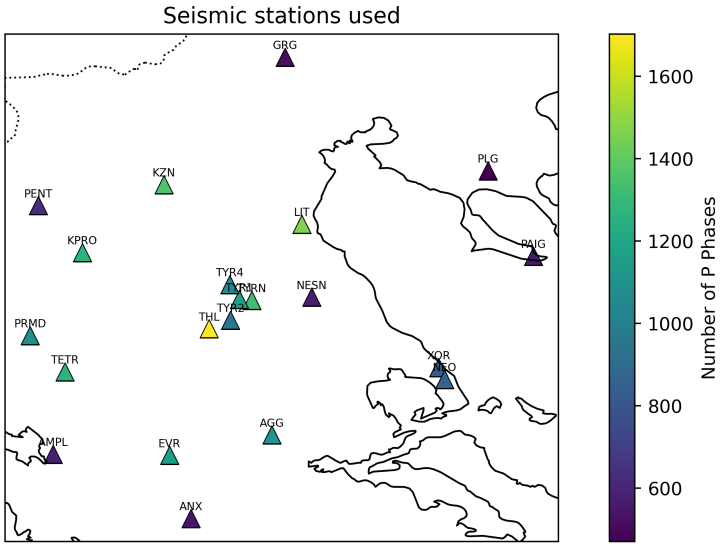
Residual threshold (|median residual|)	0.4
Stations above the threshold	PAIG, THE, LKR
Stations excluded from magnitude recomputation	PAIG, THE, LKR



Magnitude residual analysis by seismic station.

Seismic network and phases

The seismic network analysis includes **116** seismic stations. Coordinates were unavailable for **16** stations. The P- and S-phase counts for each station are stored in **Results/station_phases_counts.txt**.



Seismic stations contributing phase picks.

Phase files

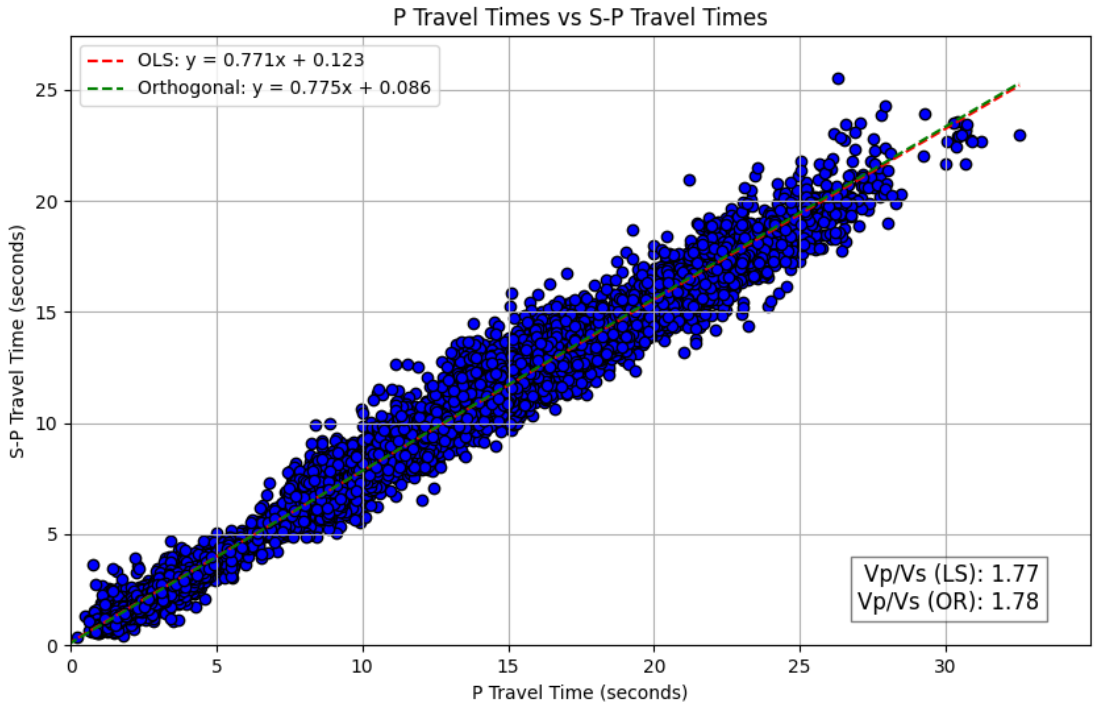
Format	File	Events	Phases
HypoDD	Results/phase.dat	1753	48015
HYPO71	Results/phases.txt	1753	47931

Station selection

Stations kept	KZN, LIT, THL
File	Results/phase_sel.dat
Events	1723

Vp/Vs estimation

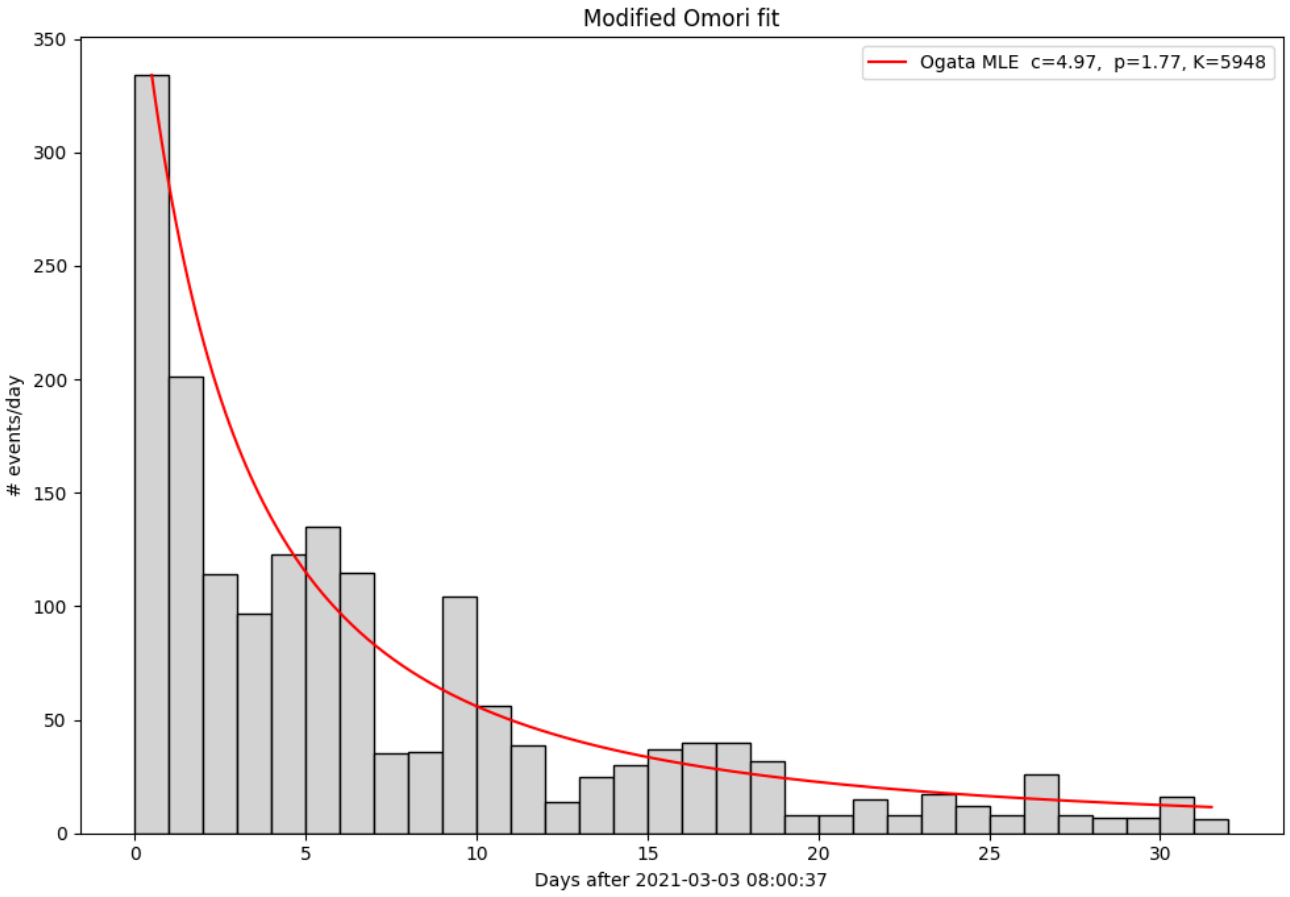
Vp/Vs - least squares	1.77
Vp/Vs - orthogonal regression	1.78



Wadati diagram and Vp/Vs estimation.

Modified Omori Law

Fit start	catalogue_start
K	5947.99
c	4.97
p	1.77



Modified Omori Law fit.

Summary

The following optional analysis results were marked as available in the analysis summary:

- Magnitude of completeness
- Magnitude analysis
- Seismic network and phases
- Phase files
- Vp/Vs estimation
- Modified Omori Law